

APSC 2113 - Engineering Analysis I
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Solutions and Explanations

Q1. How many steady states do you expect for the ODE

$$\frac{dy}{dx} = y - y^3?$$

A steady state occurs when the dependent variable does not change, so

$$\frac{dy}{dx} = 0.$$

Therefore,

$$y - y^3 = 0.$$

Factor:

$$y(1 - y^2) = 0,$$

so

$$y(1 - y)(1 + y) = 0.$$

Thus, the three steady states are

$$\boxed{y = -1, \quad y = 0, \quad y = 1}.$$

Hence, there are

$$\boxed{3}$$

steady states.

Answer: D. 3

Q2. What are the steady states for the ODE

$$\frac{dx}{dt} = 3x^2 - 2?$$

At a steady state,

$$\frac{dx}{dt} = 0.$$

Therefore,

$$3x^2 - 2 = 0.$$

Solving for x ,

$$\begin{aligned} 3x^2 &= 2, \\ x^2 &= \frac{2}{3}, \end{aligned}$$

and hence

$$x = \pm\sqrt{\frac{2}{3}}.$$

Therefore, the steady states are

$$\boxed{x = +\sqrt{\frac{2}{3}}, \quad x = -\sqrt{\frac{2}{3}}}.$$

Answer: A. $x = +\sqrt{\frac{2}{3}}, x = -\sqrt{\frac{2}{3}}$

Q3. Classify the ODE

$$A \frac{d^3 y}{dt^3} + B \frac{dy}{dt} + Cy = q(t),$$

where A , B , and C are non-zero constants and $q(t) \neq 0$.

The highest derivative present is

$$\frac{d^3 y}{dt^3},$$

so this is a **third-order** ODE.

The dependent variable y and its derivatives appear only to the first power and are not multiplied together. Therefore, the ODE is **linear**.

Because

$$q(t) \neq 0,$$

the right-hand side is nonzero. Therefore, the equation is **non-homogeneous** (it has a forcing term).

Thus, the ODE is a

third-order, linear, non-homogeneous ODE

Answer: C. Third order, Linear, Non-Homogeneous ODE

Q4. The steady state of

$$\frac{dy}{dt} = ay$$

is unstable if $a > 0$. True or False?

A steady state is obtained by setting

$$\frac{dy}{dt} = 0.$$

Thus,

$$ay = 0.$$

Since $a > 0$, we have $a \neq 0$, giving the steady state

$$y = 0.$$

To examine stability, consider a small positive perturbation from $y = 0$.

If $a > 0$, then

$$\frac{dy}{dt} = ay$$

has solutions of the form

$$y(t) = Ce^{at}.$$

Because $a > 0$,

$$e^{at} \rightarrow \infty \quad \text{as } t \rightarrow \infty.$$

Therefore, a small perturbation away from $y = 0$ grows with time rather than returning to the steady state. Hence, the steady state is **unstable**.

TRUE

Answer: A. TRUE

Q5. $x = 0$ and $x = 1$ are steady states of

$$\frac{dx}{dy} = x^2 - x.$$

True or False?

A steady state occurs when the derivative is zero:

$$\frac{dx}{dy} = 0.$$

Therefore,

$$x^2 - x = 0.$$

Factor:

$$x(x - 1) = 0.$$

This gives

$$x = 0 \quad \text{or} \quad x = 1.$$

Thus, both $x = 0$ and $x = 1$ are steady states.

Therefore, the statement is

TRUE

.

Answer: B. TRUE